

Does Social Capital Strengthen Household Resilience in Cambodia? Evidence from Panel Data

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Outline

- ➊ Motivation
- ➋ Context
- ➌ Data and Methodology
- ➍ Research Findings
- ➎ Key Takeaways

Background

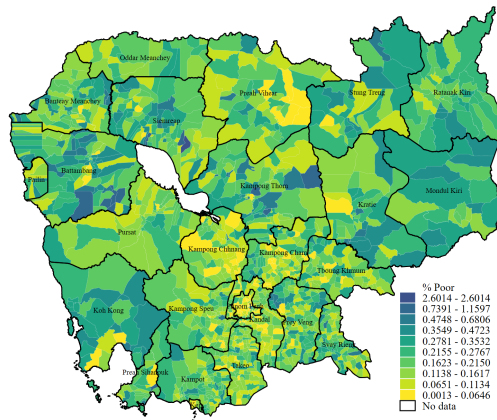
► High exposure to shocks

- People who escaped poverty remain highly vulnerable to falling back into it when exposed to economic, health or environmental shocks (CDRI 2007; WB 2013, 2022; ADB 2014).
- A significant portion of Cambodia's labour force is engaged in informal and agricultural sectors, both of which often lack adequate social safety nets (UNDP 2012; NSSF 2022).

► Resilience gaps

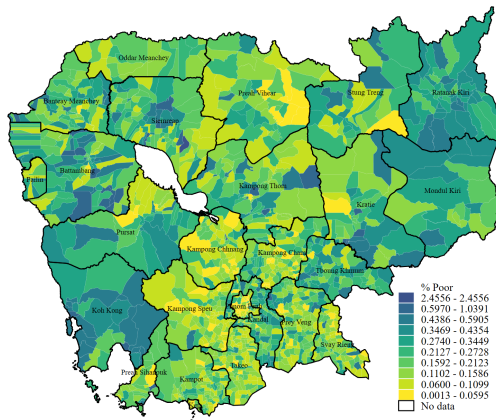
- Cambodia's social protection coverage remains limited, leaving many poor and near-poor households without a safety net in times of crisis (WB 2022; ILO 2023).
- High levels of indebtedness, particularly from microfinance loans, weaken the resilience of households, often exacerbating vulnerability when a shock hits (Iskander et al. 2023).

On average, 19.50 % of households in each commune were classified as poor.



Source: National Institute of Statistics of Cambodia

On average, 20.47 % of individuals in each commune were classified as poor.



Source: National Institute of Statistics of Cambodia

Figure: Percent of poor households and individuals both in the level 1 and 2 by communes

Motivation

- ▶ **Main Research Question:** How does social capital affect the resilience of rural households in Cambodia?
 - ① What are the structures of social capital and rural livelihoods in Cambodia?
 - ② How does social capital affect household income, consumption, and livelihood structures?
 - ③ How does social capital contribute to the development of the rural household economy?

Why does social capital matter?

- ▶ *“the features of social organisation, such as networks, norms, and trust that facilitate coordination and cooperation for mutual benefits”* (Putnam 1993, p.36)
 - It behaves in a form of formal and informal networks, and cognitive components (Putnam 1996; Uphoff 2000).
 - It can be a **horizontal relationship** (existing among equals or near-equals) or a **vertical relationship** (arising from hierarchical or unequal relations due to differences in power or resource bases) (Woolcock 2001; Grootaert and van Bastelaer 2002; Shopeju and Ojukwu 2008; Sen 2012).
 - It is a factor of **socioeconomic development**: people, institutions, and government (Bourdieu 1986; Coleman 1988; North 1990; Krishnamurthy 1999).

Context

- ▶ Social capital in Cambodia has declined since the introduction of a monetised economy during **French colonisation**, which emphasised individualism and reduced the sense of solidarity among individuals (Pellini 2005).
- ▶ It is because of the impact of the **Khmer Rouge** and **economic liberalisation** in the early 1990s that trust, social solidarity, and norms of reciprocity have declined (UNICEF 1996; Hughes 2001; Ovesen et al. 1996).
 - It is now shaped by emotional, cognitive, and historical factors (Pearson 2011).
 - Misinformation and fake news also possibly lead to low social trust.
- ▶ Today's society, people believe **no one can be fully trusted; taking care of oneself is important for survival** (UNICEF 1996; O'Leary and Meas 2001).

Context

Period	Key Characteristics of Social Capital
Pre-1975 Pre-Khmer Rouge	• Strong kinship, religious, and village-based networks (e.g., pagodas) • Community cooperation through labour exchange and ceremonial support • Informal norms governed by elders and Buddhist values
1975–1979 Khmer Rouge	• Systematic destruction of family and community bonds • Forced collectivisation and elimination of religious and traditional institutions • Breakdown of trust, widespread trauma (e.g., neighbours spied on each other)
1980s–1990s Post-Conflict	• Gradual rebuilding of trust and social ties • Emergence of local NGOs and village self-help groups • Return of religious practices and reformation of village-based institutions • Donor-led social capital interventions in reconstruction efforts
1990s Post-Conflict Recovery	• Proliferation of international NGOs and aid agencies • Start of participatory development and local governance initiatives • Fragile social cohesion, often reliant on external support
2000s Decentralisation and Development	• Boost for local participation • Expansion of community-based organisations and civil society • Stronger ties between state and community via decentralised planning
2010s Technology and Urbanisation	• Shift from traditional networks to online connectivity (social media, mobile apps) • Migration impacts rural social capital; urban informal networks emerge • Youth activism and environmental campaigns gain visibility • Evolving trust patterns in government and NGOs
2020s Resilience and Inequality	• COVID-19 highlighted community resilience through mutual aid • Informal support systems essential amid health and economic shocks • Digital solidarity and informal organising increase • Rising inequality between the poorest and richest, land disputes, and political divides challenge cohesion

Social Capital Structure

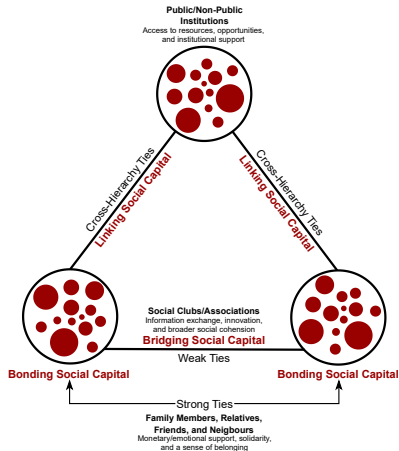


Figure: Social capital framework in Cambodia

Data

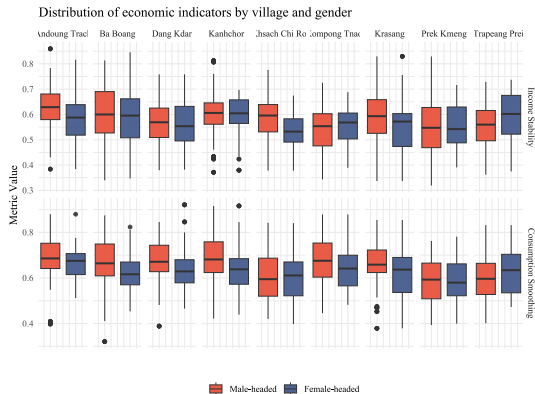
- ▶ Source: CDRI Moving Out of the Poverty Studies (MOPS) Survey 2008–2020 with 783 households across 5 waves.
- First conducted in 1993 – a unique dataset studying poverty dynamic (Murshid 1998; McAndrew 1998; Chan and Acharya 2002; World Bank 2006; CDRI 2007, 2012; Fitzgerald and So 2007; Tong 2012; Roth et al. 2017).

Sample	2008	2020	Dropped out	Attrition rate
Krasang	120	86	34	28.33%
Andoung Trach	87	61	26	29.89%
Trapeang Prei	69	54	15	21.74%
Khsach Chi Ros	121	82	39	32.23%
Dang Kdar	130	102	28	21.54%
Kompong Tnaot	123	106	17	13.82%
Prek Kmeng	120	101	19	15.83%
Kanhchor	124	98	26	20.97%
Ba Boang	128	93	35	27.34%
All villages	1,022	783	239	23.39%

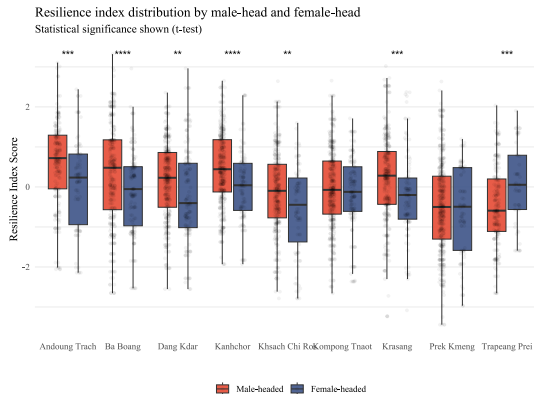
Our approach

- ▶ **Dependent:** Resilience index. We constructed a resilience index using Principal Component Analysis (PCA), based on three key dimensions: *income stability*, *consumption smoothing*, and *coping strategy index*.
- ▶ **Independent:** Social capital index (number of trust people that can provide financial support, number of group/association memberships, trust in authorities), shock, asset index, and common property resource index.
- ▶ **Control variables:** household characteristics, household identifiers, village, year, and village-by-year fixed effects.
- ▶ **Models:** Ordinary least squares and fixed-effects models.

	Mean	SD	Min	Max
Resilience index	0.000	1.090	-3.442	3.318
Consumption smoothing (1000 riels/annual)	0.644	0.105	0.321	0.921
Income stability (1000 riels/annual)	0.576	0.097	0.319	0.859
Scoping index	0.000	1.213	-0.635	7.400
Social capital index	-0.000	1.466	-2.270	5.732
Shock	0.295	0.456	0.000	1.000
Asset index	0.000	1.871	-4.922	5.827
Access to credit	0.540	0.498	0.000	1.000
CPR index	0.000	1.248	-1.452	7.221
Age of head household	53.194	12.565	16.000	95.000
Gender of head household	1.250	0.433	1.000	2.000
Years of education of the household head	3.160	3.031	0.000	18.000
Dependency ratio (%)	34.072	25.685	0.000	100.000
Agricultural land size	2.058	2.946	0.000	53.500
Real consumption per capita (1000 riels/annual)	180.116	371.414	3.425	19833.404
Real income per capita (1000 riels/annual)	704.670	1532.741	0.000	48265.184
Community/development program participation (%)	14.812	12.256	0.000	65.000
Community/development program beneficiary (%)	29.625	24.512	0.000	130.000
Real agricultural income (1000 riels/annual)	1407.332	2381.384	0.000	54636.477
Real wage (1000 riels/annual)	1767.187	4849.400	0.000	151478.906
Real remittance (1000 riels/annual)	63.101	214.680	0.000	5203.474
Number of trusted people	1.900	1.084	0.000	4.000
Number of associations	1.027	1.100	0.000	11.000
General trust perception	0.507	0.500	0.000	1.000
Real wealth (1000 riels/annual)	5866.093	19039.445	-128189.750	663630.812
Observation	3915			



(a)

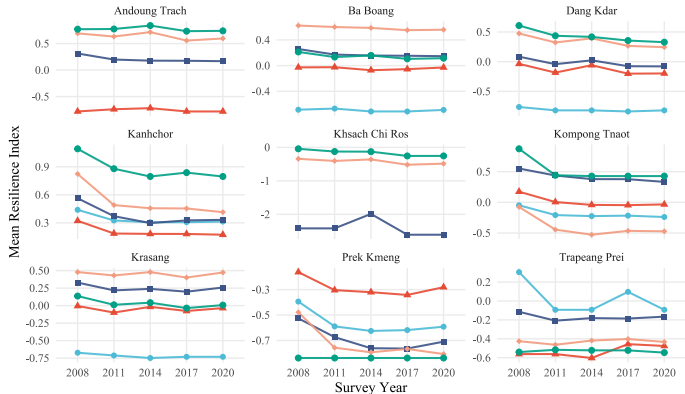


(b)

Overall, in almost all villages, male-headed households fare better than female-headed households in terms of income stability, consumption smoothing, and resilience.

Village-level resilience trends by social capital quintile

Q1 (Lowest) to Q5 (Highest) social capital households

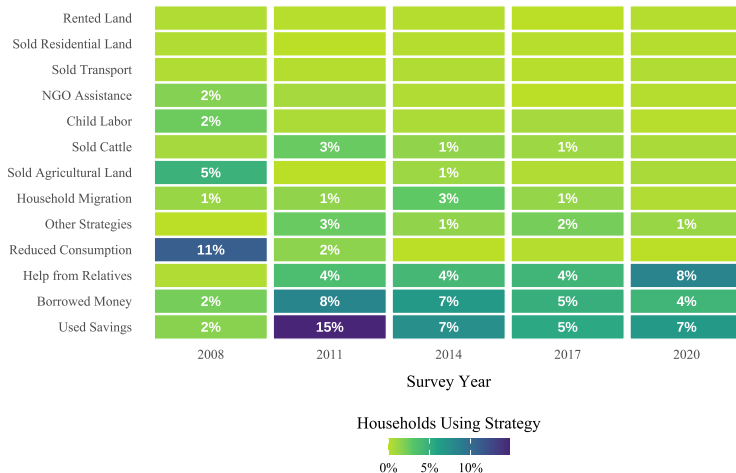


Each line represents households in the same social capital quintile within a village

The trend of resilience varies across social capital quintiles. Social capital in quintile 1 showed a low resilience index in only three villages, while social capital in quintile 5 was associated with a low resilience index in two villages.

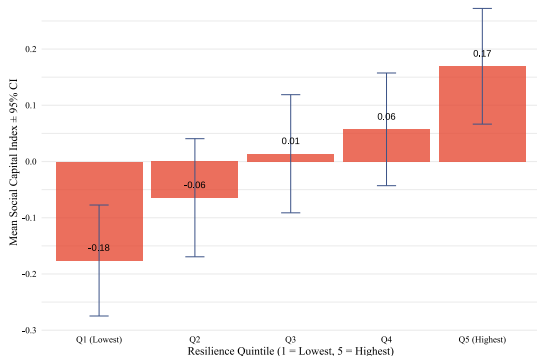
Household coping strategies

Percentage of households using each strategy by year



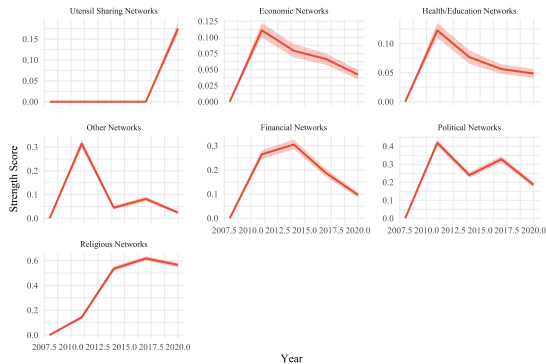
In all years, the most important coping strategies for dealing with shocks were using savings, borrowing money, and receiving support from relatives.

Social capital across resilience quintiles



(a)

Social network strength trends by category
With 95% confidence intervals

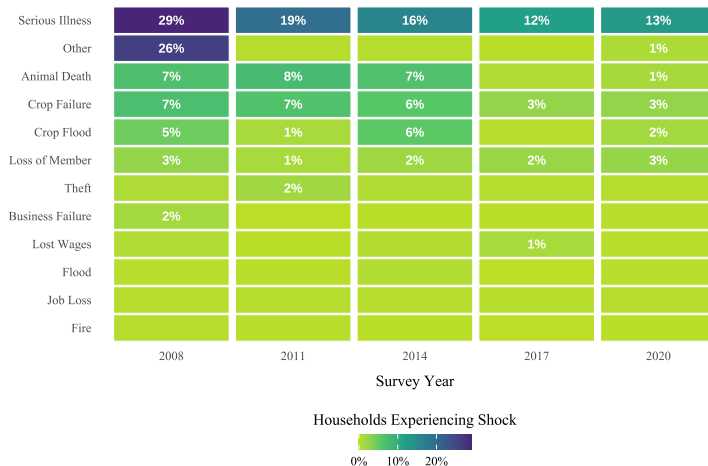


(b)

Overall, households with a high level of resilience tend to have stronger social capital networks. However, participation in groups or associations—aside from religious networks—appears to have declined.

Prevalence of household shocks

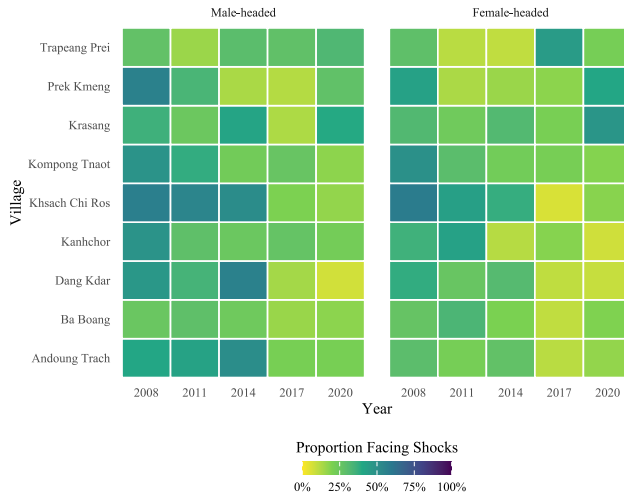
Percentage of households reporting each type of shock by year



Many households experienced shocks such as serious illness, livestock death, crop failure, flooding, and the loss of a household member (often leading to expenses for cultural or funeral events).

Shock incidence by year, village, and household gender

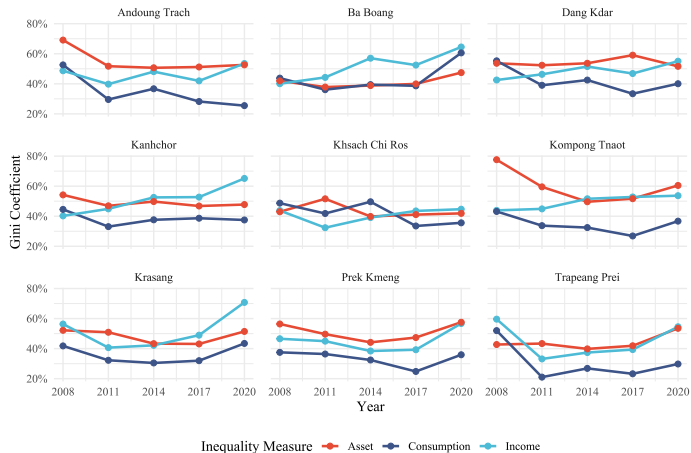
Proportion of households facing shocks



In 2008, 2011, and 2014, a larger number of households experienced shocks.

Inequality trends by village

Gini coefficient range: 0 (perfect equality) to 1 (perfect inequality)



Household asset, consumption, and income inequality by village.

Local leaders consistently earn higher trust than higher government levels



Note: Percentage shows households who answered 'Believe' out of those who gave definite answers ('Believe' or 'Not believe'). Numbers in parentheses show (count believing / total definite responses). 'Don't know' responses are excluded.

Results

Association between social capital and resilience index

Linear regression with 95% confidence interval

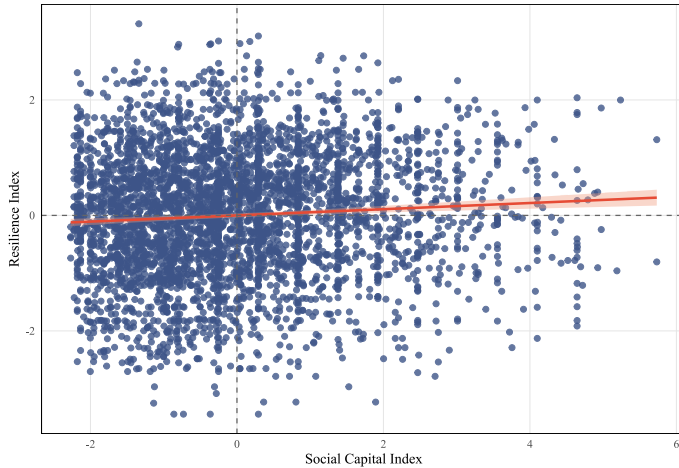
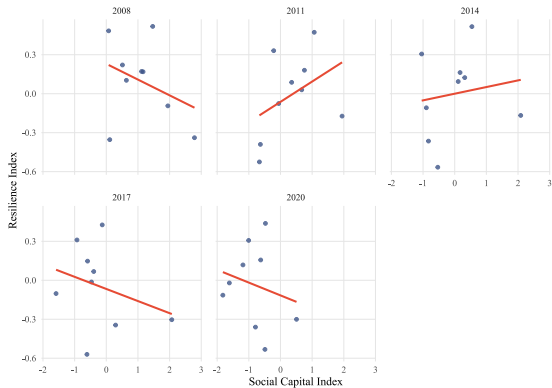


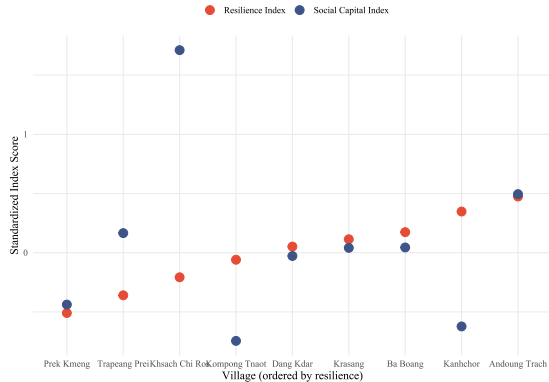
Figure: Social capital has a weak correlation with resilience

Relationship between social capital and resilience by village



(a)

Village-level averages of resilience and social capital



(b)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Social capital index	0.045*** (0.016)	0.029* (0.016)	0.034** (0.017)	0.003** (0.001)	0.003** (0.001)	0.000 (0.002)	0.002 (0.002)
Shock		0.351*** (0.041)	0.362*** (0.041)	0.275*** (0.005)	0.275*** (0.005)	0.270*** (0.004)	0.266*** (0.004)
Asset index			0.002 (0.017)	-0.015*** (0.002)	-0.015*** (0.002)	-0.003* (0.002)	-0.001 (0.002)
Access to credit			-0.112** (0.049)	-0.008** (0.004)	-0.008** (0.004)	-0.006 (0.004)	-0.009** (0.004)
CPR index			-0.005 (0.015)	0.005*** (0.002)	0.005*** (0.002)	0.002 (0.002)	0.001 (0.002)
Constant	-0.082 (0.195)	-0.184 (0.195)	-0.099 (0.199)	-0.023 (0.022)	-0.023 (0.022)	-0.058** (0.023)	-0.060** (0.023)
Household FE	No	No	No	Yes	Yes	Yes	Yes
Village FE	No	No	No	No	Yes	Yes	Yes
Year FE	No	No	No	No	No	Yes	Yes
Village-Year FE	No	No	No	No	No	No	Yes
Observations	3915	3915	3915	3915	3915	3915	3915
R-squared	0.03	0.05	0.05	0.99	0.99	0.99	1.00
F-statistic	5.16	14.09	11.09	328.95	328.28	367.07	379.14

Resilience index

- ▶ Social capital boosts resilience on average, but this effect disappears when accounting for household/village heterogeneity.
- ▶ Households experiencing shocks tend to have higher resilience. A 1-unit increase in shock exposure → 27-36% higher resilience.
- ▶ Credit may increase vulnerability (debt traps?).

	(1)	(2)	(3)	(4)
Social capital index	-63.483*** (20.278)	-63.483*** (20.299)	38.264* (20.297)	-14.664 (19.967)
Shock	-23.761 (48.930)	-23.761 (48.981)	2.594 (48.023)	-42.048 (49.069)
Asset index	186.017*** (32.163)	186.017*** (32.196)	51.911 (39.607)	34.522 (41.594)
Access to credit	-16.438 (43.482)	-16.438 (43.527)	-22.525 (42.553)	-4.702 (42.723)
CPR index	59.856*** (17.731)	59.856*** (17.749)	35.298** (16.125)	65.235*** (22.637)
Constant	478.889 (317.475)	478.889 (317.801)	1264.118*** (307.962)	1173.709*** (283.172)
Household FE	Yes	Yes	Yes	Yes
Village FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Village-Year FE	No	No	No	Yes
Observations	3915	3915	3915	3915
R-squared	0.33	0.33	0.38	0.41
F-statistic	14.68	14.65	3.84	4.20

Income stability

- ▶ Higher social capital correlates with lower income stability.
- ▶ Asset index: Wealthier households have more stable incomes.
- ▶ Shock and access to credit: Insignificant in all specifications.
- ▶ Common Property Resource (CPR): Consistently positive.

	(1)	(2)	(3)	(4)
Social capital index	2.452 (3.964)	2.452 (3.968)	15.238*** (4.910)	7.812* (4.632)
Shock	48.305*** (8.719)	48.305*** (8.728)	47.218*** (8.118)	38.514*** (8.170)
Asset index	34.373*** (5.332)	34.373*** (5.337)	28.160*** (5.706)	28.454*** (7.169)
Access to credit	26.714 (17.134)	26.714 (17.152)	29.505* (17.892)	37.342* (21.067)
CPR index	12.065***	12.065***	3.837*	3.971
Constant	130.825* (70.917)	130.825* (70.990)	213.144*** (65.184)	200.936*** (64.898)
Household Control	Yes	Yes	Yes	Yes
Village FE	No	Yes	Yes	Yes
Year FE	No	No	No	Yes
Village-Year FE	No	No	Yes	Yes
Observations	3915	3915	3915	3915
R-squared	0.26	0.26	0.29	0.30
F-statistic	16.61	16.58	14.05	12.99

Consumption smoothing

- ▶ Social capital only aids consumption smoothing when village-year fixed effects (FEs) are added.
- ▶ Shocks and assets are consistently direct role in resilience.
- ▶ Access to credit and CPR matter in certain contexts.

	(1)	(2)	(3)	(4)
Social capital index	0.025** (0.011)	0.025** (0.011)	0.001 (0.013)	0.015 (0.014)
Shock	2.093*** (0.036)	2.093*** (0.036)	2.060*** (0.033)	2.030*** (0.032)
Asset index	-0.115*** (0.013)	-0.115*** (0.013)	-0.024* (0.014)	-0.005 (0.014)
Access to credit	-0.060** (0.030)	-0.060** (0.030)	-0.043 (0.029)	-0.069** (0.029)
CPR index	0.038*** (0.013)	0.038*** (0.013)	0.015 (0.012)	0.011 (0.015)
Constant	-0.172 (0.169)	-0.172 (0.169)	-0.439** (0.177)	-0.455** (0.177)
Household FE	Yes	Yes	Yes	Yes
Village FE	No	Yes	Yes	Yes
Year FE	No	No	No	Yes
Village-Year FE	No	No	Yes	Yes
Observations	3915	3915	3915	3915
R-squared	0.74	0.74	0.76	0.77
F-statistic	328.95	328.28	367.07	379.14

Scoping strategy index

- ▶ Social Capital has conditional role in scoping strategies.
- ▶ Households facing shocks actively engage in scoping behaviors.
- ▶ Assets may not directly aid coping.
- ▶ Credit may hinder coping if it leads to debt traps.
- ▶ Common resources encourage scoping only in certain settings.

	(1)	(2)	(3)	(4)
Social capital index $\times$ Shocks	0.023*** (0.003)	0.023*** (0.003)	0.024*** (0.003)	0.023*** (0.003)
Asset index	-0.015*** (0.002)	-0.015*** (0.002)	-0.003 (0.002)	-0.000 (0.002)
Access to credit	-0.008** (0.004)	-0.008** (0.004)	-0.006* (0.004)	-0.009** (0.004)
CPR index	0.005*** (0.002)	0.005*** (0.002)	0.002 (0.002)	0.001 (0.002)
Constant	-0.022 (0.022)	-0.022 (0.022)	-0.058** (0.022)	-0.060*** (0.023)
Household FE	Yes	Yes	Yes	Yes
Village FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Village-Year FE	No	No	No	Yes
Observations	3915	3915	3915	3915
R-squared	0.99	0.99	1.00	1.00
F-statistic	311.32	310.68	345.36	357.39

Resilience index

- ▶ A 1-unit increase in social capital boosts the resilience of shock-affected households by about 2.3%.
- ▶ While credit is often promoted for resilience, its negative association here. Is it due to over-indebtedness or substitution effects?
- ▶ Wealth and common resources show conditional effects.

	(1)	(2)	(3)	(4)
Social capital index	0.025 (0.017)	0.052*** (0.015)	0.055*** (0.017)	0.063*** (0.020)
Low-income household	-0.116** (0.051)	-0.225*** (0.048)	-0.228*** (0.049)	-0.241*** (0.053)
High-income household	0.365*** (0.058)	0.284*** (0.055)	0.306*** (0.058)	0.308*** (0.060)
Shock	0.354*** (0.041)	0.357*** (0.038)	0.355*** (0.039)	0.352*** (0.041)
Asset index	0.015 (0.018)	0.000 (0.017)	0.002 (0.020)	0.002 (0.021)
Access to credit	-0.111** (0.048)	-0.067 (0.045)	-0.067 (0.045)	-0.073 (0.046)
CPR index	-0.001 (0.015)	0.010 (0.011)	0.004 (0.012)	0.006 (0.017)
Constant	-0.389* (0.211)	-0.157 (0.234)	-0.128 (0.235)	-0.174 (0.244)
Household FE	No	No	No	No
Village FE	No	Yes	Yes	Yes
Year FE	No	No	Yes	Yes
Village-Year FE	No	No	No	Yes
Observations	3915	3915	3915	3915
R-squared	0.06	0.13	0.14	0.14
F-statistic	11.34	11.27	18.13	9.96

Resilience index

- ▶ Social capital is associated with higher household resilience, especially after accounting for village-level heterogeneity (column 2 onwards).
- ▶ Low-income households: Consistently negative and significant (0.116 to 0.241), indicating lower resilience.
- ▶ High-income households: Positive and significant (0.284 to 0.365), highlighting income as a resilience driver.

	(1)	(2)	(3)	(4)	(5)	(6)
Low-income household × Social capital index	0.007** (0.003)	0.003 (0.003)	0.002 (0.003)			
High-income household × Social capital index				0.007*** (0.002)	0.005** (0.002)	0.004* (0.002)
Shock	0.274*** (0.005)	0.270*** (0.004)	0.266*** (0.004)	0.274*** (0.005)	0.270*** (0.004)	0.266*** (0.004)
Asset index	-0.014*** (0.002)	-0.003* (0.002)	-0.001 (0.002)	-0.015*** (0.002)	-0.003 (0.002)	-0.001 (0.002)
Access to credit	-0.008** (0.004)	-0.006 (0.004)	-0.009** (0.004)	-0.008** (0.004)	-0.006 (0.004)	-0.009** (0.004)
CPR index	0.005*** (0.002)	0.002 (0.002)	0.001 (0.002)	0.005*** (0.002)	0.002 (0.002)	0.001 (0.002)
Constant	-0.027 (0.022)	-0.057** (0.023)	-0.056** (0.023)	-0.027 (0.023)	-0.062*** (0.024)	-0.059** (0.024)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No	Yes	Yes
Village-Year FE	No	No	Yes	No	No	Yes
Observations	3915	3915	3915	3915	3915	3915
R-squared	0.99	0.99	1.00	0.99	0.99	1.00
F-statistic	279.54	312.92	323.08	278.21	311.10	321.54

Resilience index

- ▶ Social capital alone is insufficient for low-income resilience (e.g., may require specific village-level conditions or temporal factors).
- ▶ High-income households benefit more from social capital in building resilience. A 1-unit increase in social capital increases their resilience by 0.4% to 0.7%.

Discussion

- ▶ So, does social capital strengthen household resilience? Yes—but conditionally: only for high-income households and in villages with supportive institutions.
- ▶ Shock-exposed households exhibit higher resilience, likely due to adaptive learning and external aid.
- ▶ Endogeneity Concerns:
 - **Reverse Causality:** Resilient households may actively build social capital (rather than social capital causing resilience).
 - **Omitted Variables:** Unobserved factors (e.g., leadership, cultural norms) could drive both social capital and resilience.
- ▶ Measurement issue: We do not have qualitative data at the household level to validate the analysis.

Key takeaways

- ▶ Social capital is no longer a standalone solution to help households when declining general trust and a shift toward family-first survival strategies.
- ▶ Formal institutions are filling but not fully replacing the gap: the regression shows that **credit alone can harm resilience** if it leads to over-indebtedness.
- ▶ Resilience can no longer rely on nostalgia for village solidarity. Cambodia needs smart, adaptive systems—combining state resources, digital tools, and community trust—to face escalating shocks.
- ▶ Resilience programs must adapt to new realities: supporting vulnerable groups through both **formal and informal channels** and by going beyond shock responses to target specific areas with **indirect support**.

Thank you!

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X: @kosalnith